

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	554357
Award Program	Sustainable American Aquaculture
Project Title	Toward Responsible Pacific Bluefin Tuna Mariculture in the United States
Grant Start Date	6/29/2018
Grant End Date	6/29/2023
No Cost Extension	3/31/2023
Total Grant Budget	\$945,735.00 + 158,000.00
Total FFAR Budget	\$1,103,765
Matching Funder(s)	Ichthus Unlimited, Illinois Soybean Association, Ohio Soybean Association, Minnesota Soybean Association, United Soybean Board, USSEC, Cotton Inc.
Final Report Date	4/1/2024
Project Director/Principal Investigator Information	
Full Name	Alejandro Buentello
Email	abuentello@ichthusunlimited.com
Phone	515.333.1659
Grantee Organization Information	
Organization Name	Ichthus Unlimited, LLC
Address	109, S. 32nd Street
City, State Zip	West Des Moines IA 50265
Tax ID	47-5095923



Authorized Signing Official (ASO) Information	
Name	Alejandro Buentello
ASO Title	President
Email	abuentello@ichthusunlimited.com
Phone	515.333.1659

General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.
 - San Diego, CA Congressional district # 52nd
 - Des Moines, IA Congressional district # 3rd
 - College Station TX Congressional district # 17th
 - Mazarron, Murcia, Spain
- 1.2. How many new jobs were created because of this grant funding?
 - Mariana, Laura, Betty, Rolando, Greg, Stephenie, Diego, Omar, Vladimir, Robin, Ariadna, Daniella, Federico, Charles, Manabu, Mario, Arturo, Brian, Fernando, Alejandro, Juan, Guillermo, JimJohn, Makeena, Francisco, George, Hendrick, Kamran, Keith, Nathalee, Brooke, Harrison, and Jeannette. 33 total
- 1.3. How many jobs were maintained because of this grant funding?
 - Minimum 10 at any given time
- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.
 - No changes. The organization has been a for profit LLC from the beginning.
- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.
 - No changes.

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the



importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in United States (US) food and agriculture systems.

An increased demand for edible fish worldwide has created a need which can no longer be met by wild fisheries. However, current aquaculture practices need improvement so this industry can further expand. Bluefin tuna species (BFT) in the Atlantic and Pacific Oceans (ABFT, *Thunnus thynnus*, PBFT, *T. orientalis*) are commercially relevant and important for both aquaculture and the sports fishing industry in the US. The US seafood market is the second largest in the world but, contrary to early American history of abundance, the US now depends on imported seafood to supply more than 80% of its domestic consumption. Our seafood trade deficit now exceeds \$20 billion per year, a continuum of multi-decade deficit.

With generous support from the **Foundation for Food and Agriculture Research – American Aquaculture Program**, the USDA-SBIR Program, and the Illinois Soybean Association which, together with Ichthus Unlimited, LLC (IU) funding, the first US BFT hatchery/wet laboratory was established in San Diego, CA. From 2018 – 2023 IU conducted pioneering scombrid research aimed at controlling reproduction, mastering larvi-culture, and developing sustainable feeds for scombrid species. Marked improvements have been made to prevent tank collisions, control cannibalistic behavior, and increase overall survival. For example, at IU's hatchery survival of bluefin tuna juveniles to 1 g (~35 days post hatch) now typically exceeds 40%.

After the successful proof-of-concept this technology is now ready for commercial deployment which we expect to attain in the US Gulf of Mexico (GOM), within NOAA's central and western Aquaculture Opportunity Areas in the next few years. In collaboration with an outstanding group of multidisciplinary scientists, IU continues to make strides on early vision development, the ontogeny of regional endothermy and nutritional optimization to further improve its bluefin tuna farming technology.

The potential to utilize re-purposed offshore oil production platforms will be realized by capitalizing on technology generated by IU during the lifespan of the FFAR project. As soon as Q4 2024 it may establish the **first commercial ABFT hatchery in the US** on the lower deck of Arena Energy South Pass 83 platform (28°47.3' N, 89°14.53' W; on 470 ft of water) followed by a grow out operation in the same location. This will maximize the sustainability of ABFT farming, responding to **America's demand for domestically and responsibly grown seafood**, meeting industry needs for a thriving seafood production. The commercial culture would then satisfy an increasing portion of the market demand, while the wild population is given a better chance to rebound naturally. Finally, the sustainable cultivation of ABFT in the GOM offers one of the best opportunities to support economic growth throughout the GOM and the Eastern



US by fostering a robust aquaculture industry that may provide high-paying jobs from a resilient seafood sector. Once permitting adequately addresses regulatory requirements in CA, this model can be replicated in the US West Coast with PBFT.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

With FFAR's support, IU has developed reproduction & hatchery technology which makes possible the captive propagation of bluefin tuna species (PBFT & ABFT) in US waters. This technology involves a series of environmental controls, maturation induction protocols, larval rearing methods and specialized husbandry procedures that allow the production of juvenile tuna for cage stocking (tuna ranching) and future stock enhancement efforts for natural populations. Combined with IU's nutritional technology, which offers the only complete feed for bluefin tuna species available in the American continent and one of the best tuna feeds for the global tuna farming industry and solid-wall cage technology, also developed within the lifespan of this project, this technology has the potential to revolutionize global marine farming, providing solutions for the most critical aspects that today severely limit mariculture's potential worldwide. Because BFT species occur naturally in waters of the Exclusive Economic Zone of the US, and due to the contribution of this project to the responsible aquaculture of this species, this project has direct application in solving state, national and global fisheries problems. Vanishing PBFT wild stocks in the Pacific Ocean make the application of this technology in CA waters not only an economic priority but an ecologically urgent task as well.

IU is partnering with the Sea Grant College Program (SG, initially the TX SG) to communicate findings through SG dissemination channels. Outreach programs, workshops and publications on tuna aquaculture will be targeted to end users.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:

- a) **potential outcomes**, i.e., findings, results, or recommendations that



could impact U.S. agriculture and food systems, if used;

The potential **commercial** outcomes stemming from this work are significant. For example, outcomes from this project may create three **new American Export Products**. Specifically, BFT juveniles, compound tuna feed, and commercial size BFT for domestic and international markets. In addition, the future establishment of BFT offshore farms may also create several new jobs and economic development in coastal communities, initially around the US GOM and later in the US East and West Coasts. This is relevant because an increasing number of American livelihoods are tied to the ocean every year. Since BFT cultivation requires services from several spin-off/support industries, the economic benefits from tuna mariculture would be mainly reaped by coastal communities which comprise over 50% coastal populations. Learning from the Australian experience it is possible that BFT farming in the US may also become the most valuable American seafood industry. The Australian Department of Agriculture, Fisheries and Forestry, reports that in 2022-2023 the gross value of production for SBFT farming was \$169 million (M). Since its establishment, over 2,100 high paying jobs have been created by the SBFT farming industry around the Port Lincoln area in Southern Australia, contributing ~ **\$200 M/year** to the Southern Australian economy.

- b) **intermediate outcomes**, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth;

IU is collaborating with Blue Silo Aquaculture, a company that has secured one of the first permits to utilize retired oil platforms for aquaculture purposes in the GOM and with Ekstrom Aquaculture a long-standing (since 1980) and successful aquaculture company which grew to become the largest producer of hybrid striped bass in the world and the largest producer of red drum in the US. Together we aim at establishing the first commercial BFT hatchery in the US on the lower deck of retired oil platform SP-93 in the GOM and has initiated permitting efforts to deploy grow out cages for ABFT in waters around this platform.

- c) **end outcomes**, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

More and more American choose to live by the ocean every year. It is thus imperative to establish industries near the coastline to meet the increasing demand for good paying coastal jobs. At every location, each tuna ranch provides



~25 permanent jobs and 50+ temporary jobs, as well as providing business to several spin-off industries. Annual revenues of over \$25 M/year for a mid-size farm and the great market value that sashimi grade tuna fetches in domestic/international markets, make ranched tuna a most lucrative enterprise.

For San Diego, CA, this has paramount importance because it once was known as the "Tuna Capital of the World". Following the first tuna cannery startup (1911), the city became a world leader in commerce associated with tuna fisheries. By the 1960s, San Diego's third-largest economic enterprise would be tuna, superseded only by the US Navy and aerospace, employing over 40,000 San Diegans. After years of overfishing and the tuna embargo, this commerce has all but disappeared from Southern CA. Although technologically, the sustainable cultivation of BFT is at hand, CA also has one of the most stringent regulatory environments in the nation and its SG college program has not been particularly responsive for collaboration. In addition, the cost of land and electricity, together with difficulties in securing investment have tilted the balance – at the moment – to the GOM.

The sustainable production of BFT in the US makes sense because it is a matter of national security not to depend on imports for the supply of > 80% of our domestic seafood consumption. For the US this should be a strategic priority.

Information generated by this research has contributed to the optimization of BFT aquaculture, and adapted versions of this information may be applied to other scombrid fishes to improve production and help enhance recruitment success through future supplementation programs of wild tuna populations. The potential commercial outcomes stemming from this work may be significant. Building on the Southern bluefin tuna (SBFT, *T. maccoyii*) Australian experience, it can be speculated that bluefin tuna farming in the US may also become the most valuable American seafood industry. This, in terms of measures of economic change such as business sales by sector (e.g., ABFT farming and spinoff industries), and job creation.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

From a technical perspective, milestones for this project can be divided into a) broodfish work, b) hatchery work, and c) physiology work.

- a) **Broodfish work** progressed very slowly throughout the project due to some setbacks in the acquisition of thermal cages. These were not readily available from global vendors and the few that could be commercially acquired either did not meet resilience specifications for high energy marine environments



(AgriMarine – Canada) or were priced outside of the budgetary parameters for this project (Aquafarm, Neptune Tank, Norway). Our solution to this problem was twofold: The importation of fertile eggs from various collaborators in Japan (PBFT eggs), the Mediterranean Sea (ABFT eggs) and both Indonesia and Panama for yellowfin tuna (YFT, *T. albacares* eggs) allowed us to develop the necessary hatchery protocols. In addition, we developed a novel design for a solid-wall thermal tank that satisfactorily meets offshore requirements at a price 1/5th that of Neptune tanks. IU's solid-wall, polyethylene design uses fusion technology and high-density polyethylene (HDPE) tubing to create near cylindrical shapes that are engineered to be flexible and have suitable physical and chemical properties for fish cultivation. The insulating qualities of polyethylene and the tubes' internal diameter have significant thermal stability and will minimize temperature fluctuation inside the tank. This has the outmost relevance for mastering the maturation of PBFT in CA waters (and may solve some of this state's environmental reservations about tuna farming) and ABFT in the US Gulf of Mexico and Eastern US. Although the original design still awaits testing, preliminary engineering designs were submitted to the US patent application office (US Patent Application No. 62/940,528). A suitable initiative (perhaps the US Department of Energy) will be targeted to conduct in-depth engineering design and hydrodynamic modeling followed by prototype testing (1:10 scale), in wave basins with three-dimensional wave making capabilities, along with wind and current generators to simulate the most relevant conditions faced by offshore floating and sinking structures (wind, waves, and currents). Candidate facilities have been identified at Texas A&M University (Offshore Technology Research Center), the University of Iowa (Hydroscience and Engineering Wave Basin & Towing Tank) and the University of Miami (Hurricane Wind-Wave Hurricane Simulator).

In addition to overcoming key constraints for BFT reproduction and husbandry, this application also has tremendous potential in addressing major issues currently limiting mariculture's true potential as it prevents pathogen infestation of animals in the culture chambers, also blocking disease transmission to animals in the surrounding environment. It greatly reduces the risk of escapees and eliminates the release of uneaten feed and metabolites to the natural environment. Issues such as red tides, biofouling, chemical treatments, and lack of sustainability certifications are all resolved by this technology. Its future application solely in salmon industry has immense commercial potential. This can be considered an **additional potential outcome**.

b) **Hatchery work** was delayed at the beginning but progressed ahead of pace in 2020 – 2021, 2022 and 2023. A complex network of international suppliers of fertile eggs of PBFT, ABFT and YFT was established to allow for global shipment of fertilized tuna eggs in support of our hatchery work. Environmentally, this was not an issue because a) the three species occur naturally in the US (e.g., are endemic to US waters), b) we secured all the necessary importation permits and c) the hatchery facility is located 7-miles away from the coast, and containment systems prevent all possibilities for

accidental escapement. Even during the COVID-19 pandemic, which brought extremely challenging freight conditions, transportation protocols proved successful with after-shipment survival exceeding 80% for all three species.

An outstanding group of cooperating institutions and vendors has been formed including domestic collaborators in CA, TX, MS, LA, SC, and VA in addition to productive international partnerships in Denmark, Japan, Norway, and Spain. A system of just-in-time logistics has been established such that the timely delivery of organisms of the right developmental stage is available to serve as larval tuna feed.

The main causes of larval and early juvenile mortality have been addressed and partially solved. Tank collisions were controlled with appropriate illumination as well as limiting personnel access to rearing tanks, whereas cannibalism was controlled by always maintaining a suitable amount of prey items as well as semi-continuous sorting and size-grading of fish. Although there is still much room for improvement, this component has been successfully completed and all objectives have been attained.

c) **Physiological work** includes the study of endothermy and vision development which are essential for the survival of early juvenile specimens of PBFT in Southern California waters and ABFT in US Gulf of Mexico (GOM) waters. To hunt for prey in deep and cold water, some tunas can maintain a warm body temperature that supports their central nervous system, key organs, and eyes. However, the onset of endothermy requires a specific body mass and the development of counter current heat exchange systems. With a supplemental grant from FFAR the ontogeny of regional endothermy was studied in San Diego, CA and Mazarron, Spain. Two experimental trials, each with two groups of juvenile ABFT raised from fertilized eggs at two different water temperatures, were conducted in the latter part of 2022 and 2023 (one trial in San Diego and one in Spain).

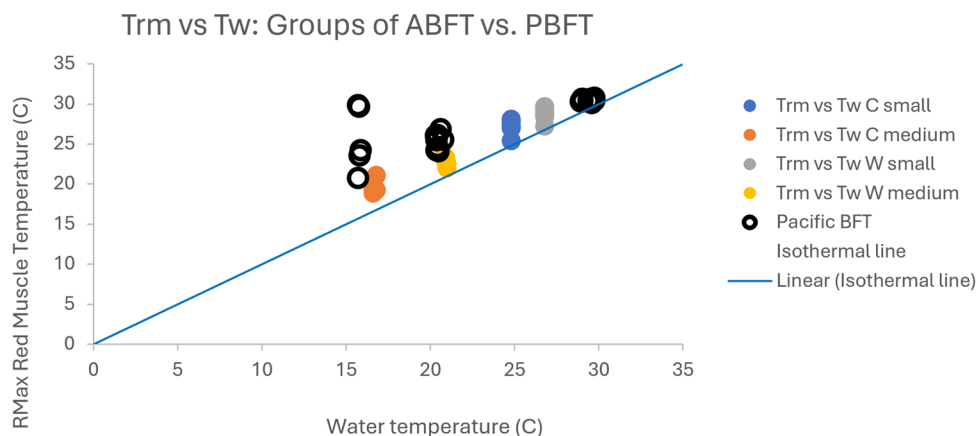


Figure 1. Red muscle temperature vs. water temperature for bluefin tuna (ABFT from Spain, PBFT from San Diego, CA, and Japan).

Acute tissue temperature measurements were recorded, and samples of the blood vessels that perfuse the red myotomal muscle and the cecal mass (warmest tissue within the viscera) and serve as vascular counter-current heat exchangers required to elevate tissue temperatures were fixed in 10% formalin for histological analyses. Sampling of the largest individuals from 2023 is still to be completed, as well as the bulk of the histological processing and analysis.

Thus far, 31 juvenile ABFT from both trials ranging in size from 21.1 cm fork length (FL) and 178 g to 45.4 cm FL and 1640 g have been sampled. The red muscle and visceral temperatures vs. water temperature, for the small and medium sized fish held in cool vs. warmer waters, are shown in Figures 1 and 2, respectively, in which data from PBFT are included for comparison.

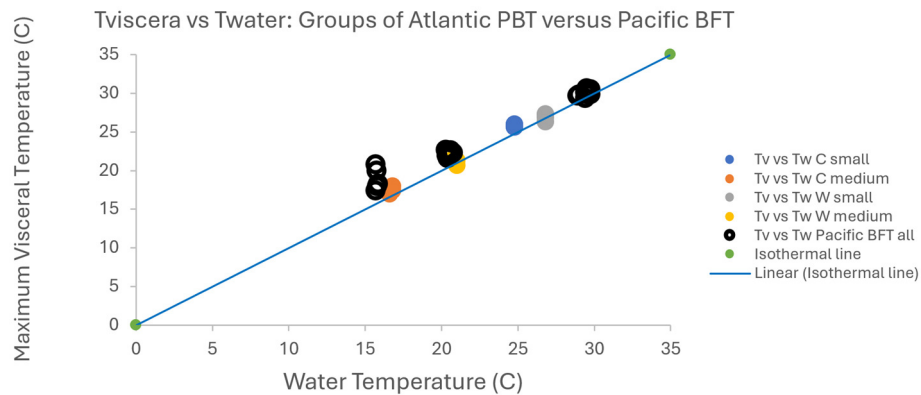


Figure 2. Visceral temperature vs. water temperature for bluefin tuna (ABFT from Spain, PBFT from San Diego, CA, and Japan).

The extent of temperature elevation ($T_x = T_{\text{tissue}} - T_{\text{water}}$) for ABFT and PBFT are shown in Figure 3 for red muscle and in Figure 4 for visceral temperatures.

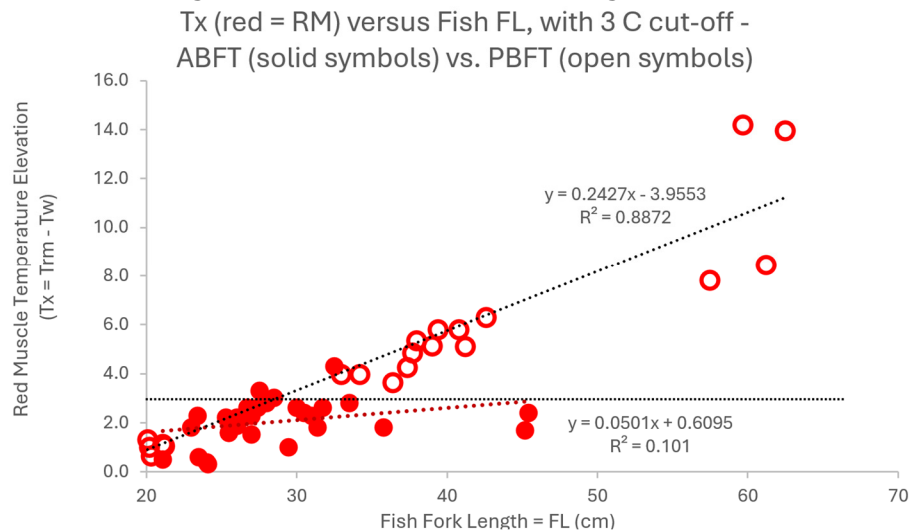


Figure 3. Red muscle temperature elevation above water temperature vs. fish fork length for bluefin tuna (ABFT from Spain, PBFT from San Diego, CA, and Japan).

Most of the ABFT fish fill in a size gap between two groups of PBFT that were previously sampled in San Diego, CA, and Japan. Red muscle temperatures from only three individual ABFT were significantly elevated (sizes 27.5, 28.5, and 32.5 cm in FL). Using the best-fit linear equation for the Tx in red muscle versus FL data for the ABFT, the size at which Tx = 3.0 C, the “cut-off” for acute temperature measurements that indicates red **muscle endothermy**, is **estimated to be 47.7 cm FL in ABFT**. This is larger than estimated for PBFT, YFT and Black Skipjack (*Euthynnus lineatus*) tunas using similar acutely measured temperature data. However, the ABFT data collected thus far are quite variable and we await additional data for larger individuals to be sampled in late 2023 or early 2024.

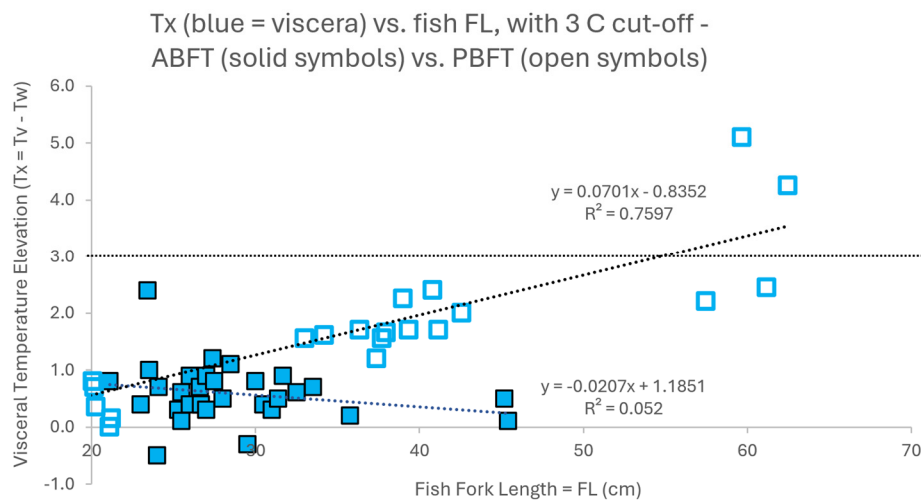


Figure 4. Visceral temperature elevation above water temperature vs. fish fork length for bluefin tuna (ABFT from Spain, PBFT from San Diego, CA, and Japan).

The negative slope (which does not differ significantly from zero) for ABFT in Figure 4 is driven primarily by one high data point. We expect a positive relationship once data for the larger fish can be included.

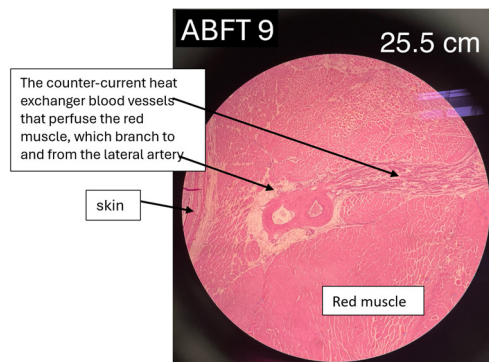


Figure 5. Histological section of lateral red muscle heat exchanger blood vessels (rete) in ABFT 25.5 cm FL. The red muscle extends to the skin and the heat exchanger is unlikely to contribute to red muscle temperature elevation at this small size, as was found for PBFT (ABFT from Spain, PBFT from San Diego, CA, and Japan).



Now, IU is pursuing an integrative approach which will provide an excellent opportunity to attain the following objectives: expedite & improve scotopic vision development – through nutritional interventions – in tuna larvae, accelerate the onset of regional endothermy – through metabolomics research – in tuna juveniles and fully understand protein synthesis and growth in ABFT – using deuterated water and microRNA to assess protein synthesis and swim tunnel studies to maximize muscle deposition. This strategy combines nutritional science and fish physiology with molecular approaches to characterize vision development, thermal adaptations, muscle growth and protein deposition.

Together with the optimization of culture environment pursued during the FFAR project as well as studies on ion-transporting cells (ionocytes), conducted in collaboration with University of California San Diego - Scripps Institution of Oceanography, data generated by this project demonstrates the feasibility of the sustainable aquaculture of bluefin tuna species in the US.

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

The lack of suitable and commercially available thermal cages forced IU to pursue an R&D effort not originally considered in this project. A suitable design and closed containment mariculture system both resilient and submersible has been developed. Although prototypes still need further testing, this technology could be applicable not only to tuna species but to other fish such as salmon, cobia, yellowtail, European seabass, and gilthead seabream. Traditional net pen cultivation (e.g., gravity cages, pods, or pyramidal cages) allows for significant interactions between wild and farmed fish, causing several negative effects including the risk of escapees and the transmission of pests and pathogens. Together with the detrimental impact on the benthonic environment directly below the cages, these issues currently limit the responsible expansion of mariculture. This expansion is indispensable to meet the nutritional needs of the world's growing population. To migrate mariculture operations to new, less sheltered environments the cage systems need to be more resilient against extreme weather conditions (e.g., storm surges). Also, to obtain environmental certification, fish farmers using net pens – be it twine or copper netting – face increasing challenges to demonstrate environmental responsibility. IU has filed a provisional patent for this technology and is presently seeking funding support to conduct the necessary prototype R&D trials.

Another aim that has changed is the target market. This has to do with the extremely stringent and slow-moving regulatory environment for ocean use of the state of CA as well as the state-wide process of gentrification. This process not only drives vulnerable diversity out of the state but also expels small companies.



As case in point, IU's experimental hatchery located at 9689 Distribution Av., San Diego CA 92121 endured 4 rental increases over the life span of the project, some of these over 20% of the previous rent. Because the technology generated has not yet deployed commercially IU decided to pivot to the US Gulf of Mexico where a much more welcoming environment has been found. With the added advantage of being able to utilize re-purposed oil platforms it appears as a strait forward decision but, we will find a way to restore the tuna industry to San Diego once we find the right investment partner.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Because the hatchery work reached a point in which replicable results could be obtained on a yearly basis, the last reporting period centered on physiological work to further improve our hatchery protocols. PBFT and ABFT were raised in San Diego, CA and in collaboration with the Spanish Institute of Oceanography and the US National Science Foundation, studies on the ontogenesis of endothermy in these species were concurrently pursued.

Tunas can elevate their red (slow-twitch, oxidative) skeletal muscle, visceral and cranial temperatures significantly above the ambient water temperature with the aid of specialized blood vessels – retia mirabilia – that conserves metabolic heat. The ontogeny of this phenomenon, known as regional endothermy, has been studied in young PBFT (by members of the IU team) but not on ABFT.

Tunas can be classified as either 'temperate' or 'tropical'. The 'temperate' tunas spend significant periods of time in cool waters ($< 15^{\circ}\text{C}$) at higher latitude or below the thermocline, whereas the 'tropical' tunas are restricted to warmer waters ($> 20^{\circ}\text{C}$). Regardless of whether they are 'temperate' or 'tropical', all tunas return to warm tropical waters ($\geq 24^{\circ}\text{C}$) to spawn and their larval distribution reflects this fact. Temperate tunas include albacore tuna (*T. alalunga*) and the bluefin tunas, PBFT, ABFT and SBFT. Temperate tunas exhibit more extensive regional endothermy than tropical species and this, along with their larger body sizes, presumably contributes to their ability to spend long periods of time in cooler waters. For the success of hatchery propagation efforts, there may be a minimum size at which temperate tuna juveniles are able to withstand low water temperature conditions.



The quantification of peritoneal temperature and heart rates of endothermic organisms are essential to comprehend how “temperate” tunas respond to environmental variability and to evaluate their sensitivity to environmental change. Commercially available implantable heart rate & internal temperature bio-loggers provide the opportunity to study juvenile ABFT in vivo.

Specific objectives

- I.** Characterize, for the first time, the ontogeny of regional endothermy in ABFT using hatchery raised fish and contrast to wild and hatchery raised PBFT
- II.** Determine the applicability of an implantable device, using ventrolateral incisions for near-heart placement, and
- III.** Conduct metabolomics sampling to assess the effects of environmental temperature on ABFT metabolism and relate this to survival and potential recruitment success

Major activities

Temperature regimes: Two partial recirculation systems were used in Spain and two in San Diego, each with independent biofilters and temperature control. We manipulated temperature regimes within ranges occurring in nursery grounds in the Mediterranean Sea (~20 and 28 C) for ABFT and between Ryukyu Island (Japan) and northeast Taiwan (~21 and 29 C) for PBFT. Salinity, pH, and dissolved oxygen followed conditions reported for spawning and nursery locations for both species. The effects of temperature regimes were evaluated on juvenile tuna of ~300, 700 and 1,500 g which required different time intervals due to the influence of temperature on tuna growth. For the 27 C regime, 250 juvenile fish (~5 g) were stocked by the first week of Aug. After ~40 days fish reached ~300 g at which point 6 animals ($n = 6$) were sampled for tissues. At this sampling interval, three individuals on each temperature regime were fitted with an implantable heart rate activity and temperature logger (Star Oddi DST milli-HRT ACT) via ventrolateral incision followed by a suture wrap. Logger-fitted fish and the remainder fish in this tank were transferred to larger sized tanks. Due to the rapid growth supported by near optimal temperature, these animals reached 700 g in about 40 days more (early Oct). Six more animals were sampled for tissues (metabolomics) at this time. Finally, by the first week of Dec animals reached ~1,500 g (targeted size) and the final tissue collection took place.

Similarly, for the 20 C thermal regime, the stocking date and density was as described above. However, the lower temperature made it necessary to keep the fish probably past Nov to reach the first two sampling points (300 and 700 g). At this point fish were transferred to larger size tanks until the target size was attained (Dec – Jan).

Implants: Tuna of each size was anaesthetized prior to surgery with 2-phenoxyethanol at a dose of 0.2 mL/L until the loss of equilibrium, upon which they were placed in an operating sling lateral side up. A soft wet microfiber cloth



covered the eyes while anesthetic water was be poured over the gills and cloth to maintain anesthesia and keep external organs moist. A single ventrolateral incision was practiced immediately posterior of the pectoral girdle of ~25 mm. The loggers were placed within the body cavity posterior to, but in contact with, the pericardial membrane, while carefully avoiding membrane piercing. Two sutures were attached to the logger which was inserted with the transparent-epoxied part, where the temperature sensor is located, pointing posteriorly. Following surgery fish were released back into the culture tanks with conspecifics. Recovery time was monitored, and fish was released once it regained equilibrium and showed signs of vigor.

Ontogeny of regional endothermy in juvenile bluefin tuna: Assessment of the ontogeny of red muscle and visceral retia – blood vessels that serve as counter-current heat exchangers required for endothermy – were evaluated with increasing fish size and the capacity for heat retention, thus endothermic ability was measured as follows: Water temperature regimes of 20 and 27 C were imposed on tuna as previously indicated. At the first sampling interval thermoregulation was documented on six fish per thermal treatment. At the next sampling interval, fish were sampled again ($n = 6$, per size class). To corroborate implant readings temperatures were immediately measured in muscle, viscera and cranium using a thermocouple and digital thermometer.

Histological analyses of the red muscle and visceral retia: After euthanasia (MS-222, 400 ppm, 3 min) and spike to the brain, the same sampled fish were used for red muscle collection (left side, skin attached for orientation) and visceral retia (including liver and caecal mass), placed in labeled cassettes, fixed in 10% (v/v) neutral buffered formalin and transferred to our collaborating lab (California State University – Fullerton) for paraffin histology processing. Stained tissue sections were microscopically examined, images captured on Q Capture Pro software and measurements of rete size obtained with ImageJ software.

For the red muscle retia both the length of the rete blood vessels and the number of rows of blood vessels comprising it were measured, and for the visceral rete, total cross-sectional area was measured.

Quantification of red muscle distribution and total mass of red muscle: Total body mass was determined using a spring balance immediately after the fish was euthanized. Fish was sectioned with a band saw, sections photographed, and ImageJ used to calculate the total area of the cross-section and the area of the cross-section occupied by the red muscle. The total mass of red muscle was calculated by adding the volumes in each section along the body and multiplying by the published density value for tuna muscle to give a mass value for each fish. Student's t-test will be used to test whether a given mass-scaling coefficient (from log-log plots of red muscle mass versus fish mass) differs significantly from a specific value.



Metabolomics and thermal adaptation: Nuclear Magnetic Resonance (NMR)-based metabolomic techniques will be used to assess the effect of differential temperature regimes on early juvenile ABFT through our collaboration with Dr. Valerie Copie (Montana State University). Blood was obtained from juvenile fish (in 2 mL tubes), centrifuged, separated into plasma and blood pellet, and immediately flash frozen (FF) in liquid nitrogen. After temperature measurements, fish were dissected and gills, liver, heart, gallbladder, spleen, and intestine obtained. Also, left dorsal loin muscle tissue was collected for metabolomic analysis. All tissue samples were placed in individual, pre-labelled cryovials, FF and processed. After consolidation of samples Dr. Copie has provided prices estimates for processing these samples. Once funding is secured for these tasks and the samples analyzed, an addendum to this final report will be sent to FFAR.

As stated in section 2.4, these studies indicate that red muscle endothermy is attained in hatchery raised ABFT and PBFT at about 47.7 cm FL. This size is larger than that estimated for wild caught PBFT, YFT and black skipjack tunas using similar acutely measured temperature data. Hatchery fish appears to have a somewhat delayed regional endothermy. Metabolomic data on both ABFT and PBFT, both wild caught and hatchery raised should help elucidate factors causing this delay.

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.
Include approaches taken to ensure robust and unbiased results.

The conducted research work is a multi-institution collaboration between IU, Texas A&M University, the Spanish Institute of Oceanography and more recently, NOAA-Southwest Fisheries Science Center, California State University Fullerton, the National Science Foundation, the University of California San Diego – Scripps Institution of Oceanography and Montana State University. By pursuing this research jointly, scientific rigor, objectivity and verification is warranted. Also, members of the research team are at the forefront of scombrid science and have a strong track record of scientific productivity and publication in peer-reviewed journals. The scrutiny of external reviewers also assures a sound scientific approach and data interpretation.

- 2.8. List key stakeholders that could be served by results of this project.

With NOAAs' recently introduced initiative of Aquaculture Opportunity Areas future American tuna farmers operating in the chosen areas will directly benefit from this research. Likewise, sport fisherman operating in Southern California and the US Gulf of Mexico will likely observe spikes in PBFT and ABFT catches as a direct consequence of this research. Scientists in the above-mentioned academic institutions and others are benefiting from access to this unique species. Private organizations such as Ekstrom Aquaculture, Blue Silo Aquaculture and others could



become end users of this technology. Finally, the public at large will benefit in terms of job creation and economic development in Southern California and the US Gulf of Mexico.

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

At least 3 press releases have been published in specialized outlets describing the results of this research. A publication entitled "Bluefin tuna hatchery success: A first step in resurrecting San Diego, California as the "Tuna Capital of the World"" (HF&M 2020-8-4-8-11-Bringing back bluefin tuna to SD – Michelato et al) made the cover of a recent issue of the Hatchery & Feed Management Magazine.

Invited oral presentations were delivered at the World Aquaculture Society Meetings in New Orleans, LA (2019), Honolulu, HI (2020), and San Diego, CA (2022), as well as at the European Aquaculture Society meeting in Rimini, Italy (2022).

Documentary videos on works from this project are posted on IU's website (<https://ichthusunlimited.com/>) in the news and media section. Through our collaborations with the Texas Sea Grant College Program (Laura Picariello) dissemination channels, proven effective already will be used to further disperse this information among communities of interest. Articles in Texas Shores magazine and the Texas Sea Grant quarterly Fisheries, Aquaculture and Seafood newsletter (Get Hooked) will present results of this work.

- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

The most important hurdle to deploy this technology commercially is securing



investment and is common to other small businesses. IU has been in survival mode since the COVID 19 pandemic but is still delivering results. We did not deploy thermal cages at our concession site off the Coronado Island. Norway-based Aquafarm was not able to deliver these cages in San Diego. We did sign an agreement with Trojan Pipe, a US-based company (TX) which could help us build and deploy these cages. However, oil-related revenue for Trojan Pipe dried up during the pandemic and they were not able to provide the committed investment. We have explored options through the Small Business Administration of the Federal Government but - even though this is supposed to be a government backed loan program - the lending is conducted by private banking institutions that require guarantees we cannot provide. We have hired a new capital raise advisor and are currently pursuing new investment opportunities.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

This year M.S. student Jeannette Bush (California State University Fullerton) travelled to Mazarron, Murcia in Spain to participate in various samplings of the endothermy component of this project. She learned dissection techniques as well as application of anesthesia to ABFT, use of thermocouples to secure skeletal muscle readings, surgical procedures for the implantation of data loggers and histology procedures. The analyses of these tissues as well as those originating at IU's hatchery in San Diego, CA will conform Jeannette's Master's thesis. Spanish students in Mazarron Edurne Blanco, Patricia Reglero, Pilar Baixauli and Amanda Miras received similar training.

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

- Undergraduate: (5) George Nassar, Hendrick Enciso, Kamran Walsh, and Brooke Penny, Harrison Gilroy
- Graduate: (4) Laura Martinez, Daniella Acevedo, Nathalee Ulloa,



Jeannette Bush

- Postdoctoral: (2) Mariana Michelato and Juan Sierra

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

At least 3 press releases (Undercurrent News, Hatchery International, Global Aquaculture Advocate) have been broadcasted in specialized outlets describing results of this research. A publication entitled "Bluefin tuna hatchery success: A first step in resurrecting San Diego, California as the "Tuna Capital of the World"" (HF&M 2020-8-4-8-11-Bringing back bluefin tuna to SD – Michelato et al) made the cover for the recent issue of Hatchery & Feed Management Magazine. Symposium Proceedings of the World Aquaculture Society meeting and the European Aquaculture Society meeting have abstracts of oral presentations that were delivered with results of this work.

- 3.2. Please provide a list of citations for the information products produced during the project.

HF&M 2020-8-4-8-11-Bringing back bluefin tuna to SD – Michelato et al. 2020

Hatchery International 2-8-2021-US farm ramps up bluefin tuna development-Waycott 2021

Global aquaculture advocate 10-12-2020-A look inside CA's new bluefin tuna hatchery-Leschin-Hoar 2020

Buentello, A., Acevedo, D., Sierra, J., Bearden, D.W., Casu, F. 2019. From baitfish feeding to balanced nutrition: A leap forward in sustainable tuna farming. Proceedings of the World Aquaculture Society Meeting, New Orleans, LA, Mar 7-11.

Casu, F., Watson, A. Buentello, A., Bearden, D.W., Sierra, J., Acevedo, D., Denson, M.R. 2020. NMR-based metabolomic evaluation of a formulated feed in juvenile yellowfin tuna, *Thunnus albacares*. Proceedings of the World Aquaculture Society Meeting, Honolulu, HI, Feb 9-12.

Seoka, M., Martinez Steele, L., Michelato Kawakami, M., Buentello, A. 2022.

Recent advances in tuna hatchery procedures. Proceedings of the World Aquaculture Society Meeting, San Diego, CA, Feb 28 – Mar 4

Buentello, A. Muñoz, V., Gamas, P., Acevedo, D., Reguera, M. 2022. First successful Atlantic bluefin tuna (*Thunnus thynnus*) feeding demonstration with formulated diets: the only sustainable future of tuna farming. Proceedings of the European Aquaculture Society Meeting, Rimini, Italy, Sep 27-20

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Nothing to report yet. Scientific articles targeting peer reviewed journals are being composed and will be submitted in high impact aquaculture publications.

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

<https://ichthusunlimited.com/>

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#1	#	Patent No. 62/940,528
Copyrights	#	#	
Trademarks	#	#	
Inventions	#1	#	

- 3.6. Are any of the information products produced during this grant that are



confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

YES

- A closed containment system for the sustainable cultivation of marine fish in open ocean conditions – we have submitted a patent application and plan to make a business around this technology
- A device to prevent tank collisions in early juvenile tuna – this source of mortality is a major issue in tuna culture and our solution is not available anywhere else. As a small business we need a competitive advantage that would allow our business to become established

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

We interact with several academic institutions throughout the US (researchers) and collaborate closely with NOAA's Southwest Fisheries Science Center (policy makers). Through established channels from Birch Aquarium (also a collaborator) we make this information available to the public. Outreach programs, workshops and publications on tuna aquaculture are targeted to end users. A competitive advantage of this project is that it has been using dissemination channels which have been proven effective for personnel training. Collaborator Laura Picariello (Texas Sea Grant – Extension Specialist) has substantial experience in marine fisheries and aquaculture, extension, and information transfer.

4. Data Management

4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

Selected data will be made publicly available in the form of scientific presentations at the end of the project. Due to facility schedules at collaborating institutions some of the data loggers are still in the peritoneal cavity of fish. It is expected these can be retrieved at the end of summer 2024. Once data is downloaded it will be used to structure manuscripts. As the peer-review process for resulting



publications lays outside the research team's control, a period of two years after the end date of the grant will be considered as end point (Dec 2025). In addition, data will be visible and available starting Jan 2026. The resulting papers and other publications will be maintained in IU's Institutional Repository and made available free of charge as PDF downloads from IU's website. FFAR will be identified as funding source in all resulting scholarly publications including a reference to the American Aquaculture Program and specific award number (554357). Data sharing includes dissemination outlets described before.

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

Nothing to report.

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

Nothing to report.



Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Alejandro Buentello

PI Signature: A. Buentello Date: 3/31/2024

Authorized Signing Official (ASO) Name: Alejandro Buentello

ASO Signature: A. Buentello Date: 3/31/2024